A Physics-Informed Fourier Neural Operator for Rapid Emulation of 1D Dynamic Earthquake Rupture

Aimran

Department of Mathematical Sciences
New Mexico State University

NMSU–UTEP Joint Mathematics Workshop
April 2026

Outline

1. Motivation
3. FNO Architecture
4. Dataset & Training
5. Results
6. PINO Extension
7. Conclusions & Future Work

∂s
b$\partial t = \mu \frac{\partial v}{\partial x}$(constitutive / Hooke's law)

Spatial discretization: Summation-by-Parts (SBP) finite differences

- 4th-order interior, 2nd-order boundary stencils
- Satisfies a discrete energy estimate: $\frac{d}{dt}\|u_h\|_E^2 \leq 0$

Boundary conditions: Simultaneous-Approximation-Terms (SAT)

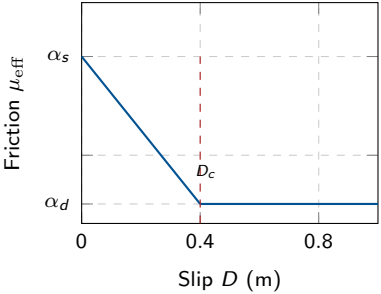
- Fault interface: impose traction continuity + friction law weakly
- Absorbing (outflow) conditions at outer boundaries $x = 0$, $x = 2L$

Fault Friction: Slip-Weakening Law

Fault traction evolves with **total slip** D:

$$\tau_{\mathrm{SW}}(D) = \sigma_n \left[\alpha_s - (\alpha_s - \alpha_d) \min\left(\frac{D}{D_c}, 1 \right) \right]$$

- $\tau = \tau_0 + s_l(x_f, t)$ (traction from left)
- Rupture nucleates when $\tau \geq \tau_{\mathrm{SW}}$
- Friction drops from $\alpha_s \sigma_n$ to $\alpha_d \sigma_n$ over distance D_c



Free parameters (our operator input):

Symbol	Description	Range
τ_0	Background stress	[78, 88] MPa
α	Static friction	[0.62, 0.74]

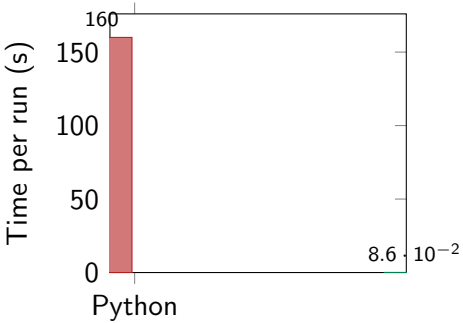
Time Integration & Numba Acceleration

Time integration

- Classical 4th-order Runge-Kutta (RK4)
- CFL-stable time step:
  $\Delta t = \mathrm{CFL} \cdot \Delta x / c_s$
- $n_x = 501$ grid points, $n_t \approx 577$ steps
- Fields stored every `iplot` steps

Numba JIT acceleration

- All kernels: `@njit(parallel=True)`
- Pure-Python serial: **160 s/run**
- Numba (1 thread): **0.086 s/run**
- **1868×** speedup
- Multi-thread: *negative* scaling —



1868× faster than Python serial

Operator Learning: Problem Formulation

Goal: Learn the parameter-to-solution operator

$$\mathcal{G}_\theta \;:\; \mathbf{p} = (\tau_0,\, \alpha_s,\, \alpha_d,\, D_c) \;\longmapsto\; \big(v_l(x,t),\; s_l(x,t),\; v_r(x,t),\; s_r(x,t)\big)$$

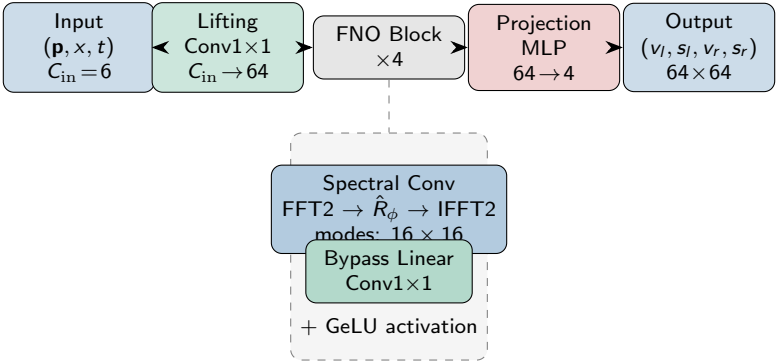
Why not a standard neural network?

- Output is a *function* on a 2D grid $(x,t) \in [0,L] \times [0,T]$
- $64 \times 64 \times 4 = 16{,}384$ output values
- Solution structure is *non-local* (wave propagation)
- Standard CNN/MLP won't generalize across resolutions

Why Fourier Neural Operators (FNO)?

- Learns *integral operators* in Fourier space
- Resolution-invariant by design
- Exact global basis functions (Fourier modes)
- State-of-the-art on PDE solution operators
- Efficient: $\mathcal{O}(n \log n)$ via FFT

FNO-2D Architecture



Input: $(B, 6, 64, 64)$
params + (x, t) coords

FNO Block: spectral conv
+ pointwise linear + GeLU

Total params:
8.4 million

Spectral Convolution Layer

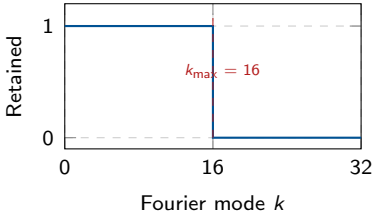
Key idea: replace spatial convolution with a *global* Fourier integral operator

$$(\mathcal{K}_\phi\, u)(x,t) = \mathcal{F}^{-1}\Big[\hat{R}_\phi(k_x,k_t)\cdot\mathcal{F}[u](k_x,k_t)\Big](x,t) \tag{1}$$

Implementation

1. **FFT2** the input: $\hat{u} = \mathcal{F}[u]$
2. Multiply by complex weights $\hat{R}_\phi$ for the first $k_{\max}$ modes
3. **IFFT2** back to physical space
4. Add pointwise linear bypass

$\Rightarrow$ Modes beyond $k_{\max}$ are *truncated* (low-pass filter)



Dataset Generation via Latin Hypercube Sampling

Sampling strategy

- Latin Hypercube Sampling (LHS) over 4D parameter space
- LHS ensures uniform coverage — no clustering
- Validity constraints enforced:
  - $\alpha_d < \alpha_s$ (velocity-weakening)
  - $\tau_0 > \alpha_s \sigma_n$ (nucleation condition)

Dataset splits

Split	Samples	Time
Train	200	∼46 s
Val	50	∼12 s
Test	50	∼12 s

Output per sample

Downsample simulator output to 64×64 grid:

- $v_l(x,t)$, $s_l(x,t)$ left domain
- $v_r(x,t)$, $s_r(x,t)$ right domain
- Fault time series: slip, slip-rate, traction

Normalization

- Parameters: zero-mean unit-std (fit on train)
- Fields: channel-wise zero-mean

Training Setup

Loss function (supervised)

$$\mathcal{L}_{\text{data}} = \frac{1}{N} \sum_{i=1}^{N} \frac{\|\mathcal{G}_\theta(\mathbf{p}_i) - u_i\|_2}{\|u_i\|_2}$$

Relative L^2: scale-invariant, treats all fields equally.

Optimizer: AdamW

- $\ell_r = 10^{-3}$, weight decay $= 10^{-4}$
- Gradient clip at norm $= 1.0$

LR Schedule: cosine decay

- Linear warmup for first 10 epochs
- Cosine anneal: $10^{-3} \to 0$ over 200

Hardware & settings

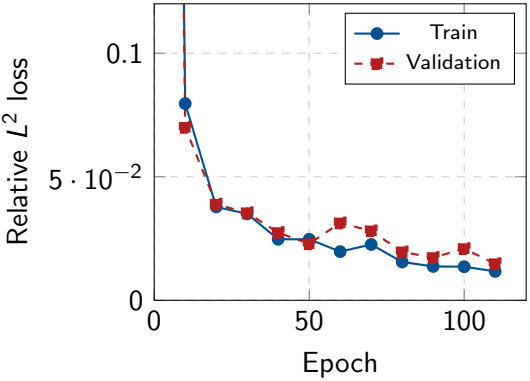
- CPU-only (no GPU on this node)
- Batch size: 16
- Epochs: 200
- ≈14.4 s/epoch on CPU
- Total training time: ≈48 min

Future: PINO loss

$$\mathcal{L} = \mathcal{L}_{\text{data}} + \lambda_{\text{PDE}} \mathcal{L}_{\text{PDE}} + \lambda_{\text{BC}} \mathcal{L}_{\text{BC}}$$

- Currently: $\lambda_{\text{PDE}} = 0$ (baseline)
- PDE residuals computed *exactly* in

Training Convergence



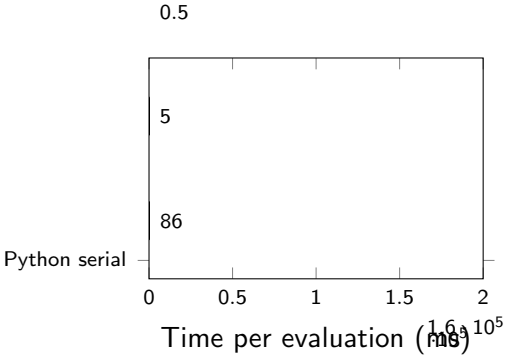
Key observations

- Epoch 1 $\rightarrow$ 10: **92% drop** in loss (network learns dominant modes quickly)
- Train $\approx$ Val $\Rightarrow$ no overfitting
- **Best val loss: 0.0148** at epoch 110 ($\approx$1.5% rel. error)
- Still improving — training ongoing

Summary so far

FNO achieves <2% relative L^2 error on all 4 fields after 110 epochs of CPU training.

Expected Speedup at Inference



Method	Time	Speedup
Python serial	160 s	1×
Numba (np=1)	86 ms	1,868×
FNO (CPU, est.)	∼5 ms	∼32,000×
FNO (GPU, proj.)	<1 ms	$>10^5\times$

Key advantage:
FNO inference ≈ matrix multiply + FFT — highly parallelizable.

Simulator must march forward in time; FNO predicts the *entire* space-time solution in one forward pass.

Physics-Informed Extension (PINO)

Idea: supplement data loss with PDE residuals — enforce governing equations in the prediction.

PDE residuals (evaluated *exactly* in Fourier space):

$$\mathcal{L}_{\mathrm{PDE}} = \underbrace{\left\| \partial_t v_l^\theta - \tfrac{1}{\rho} \partial_x s_l^\theta \right\|^2}_{\text{momentum}} + \underbrace{\left\| \partial_t s_l^\theta - \mu\, \partial_x v_l^\theta \right\|^2}_{\text{constitutive}} + \text{(right domain)}$$

Derivatives via FFT: $\hat{\partial}_x u = i k_x \hat{u}, \quad \hat{\partial}_t u = i\omega \hat{u}$

Fault boundary loss (SW friction residual):

$$\mathcal{L}_{\mathrm{BC}} = \left\| \tau^\theta - \tau_{\mathrm{SW}}(D^\theta, \mathbf{p}) \right\|^2 + \left\| s_l^\theta(x_f) + s_r^\theta(x_f) \right\|^2$$

PINO advantage over PINN

PINN uses finite differences for derivatives (inaccurate, slow). PINO uses FFT derivatives — *spectrally accurate* and fast. No spatial collocation points needed.

Conclusions

What we built

- High-performance 1D rupture simulator (Numba, 1868$\times$ speedup)
- FNO-2D surrogate: maps 4 scalar parameters to full space-time fields
- PINO extension: PDE + BC residual losses in Fourier space
- Complete pipeline: LHS data generation $\rightarrow$ training $\rightarrow$ evaluation

Results (POC)

- 300 training samples generated in $<$2 min
- Val loss $\approx$**1.5%** rel. L^2 error in 110 epochs
- No overfitting (train $\approx$ val)
- Expected speedup: $\sim 10^4$–$10^5\times$ vs. simulator
- 8.4M parameter model trains in $<$1 hr on CPU

Future work

- Enable PINO loss ($\lambda_{\mathrm{PDE}} > 0$) and measure physics consistency improvement
- Scale to 2000 samples, expand to 6-parameter space (add σ_n, c_s)
- CPU training for sub-millisecond inference

Thank You

Questions?

Repository & Contacts

Code: `/FNO/waveqlab_SF/1d_numba_PINO_FNO/`

Simulator $\rightarrow$ Numba JIT $\rightarrow$ FNO-2D $\rightarrow$ PINO

Simulator	SBP-SAT + RK4, SW friction, Numba
Surrogate	FNO-2D, 8.4M params, 64×64 grid
Current acc.	$\sim$1.5% relative L^2 (110 epochs)
Speedup	$\sim10^4\times$ (CPU), $>10^5\times$ (GPU)

SBP-SAT Discretization Details

SBP derivative operator: $D_1 = H^{-1}Q$

- H — positive definite diagonal norm matrix
- Q — nearly skew-symmetric: $Q + Q^T = \mathrm{diag}(-1, 0, \ldots, 0, 1)$
- Guarantees: $u^T H D_1 u =$ boundary terms only (mimics integration by parts)

SAT penalty for fault interface: At $x = x_f$:

$$\mathrm{SAT} = H^{-1}\left[\tau - \tau_{\mathrm{SW}}(D)\right] \mathbf{e}_{n_x}$$

where $\mathbf{e}_{n_x}$ is the unit vector at the fault node. This imposes the friction law weakly without modifying the interior stencil.

Energy stability: The SBP-SAT combination satisfies

$$\frac{d}{dt}\|u_h\|_H^2 = \text{boundary terms} \leq 0$$

ensuring the scheme is non-growing (analogous to the continuous energy estimate).

Parameter Space & Validity Constraints

All parameters

Symbol	Description	Range	Fixed?
τ_0	Background stress	[78, 88] MPa	No
α_s	Static friction	[0.62, 0.74]	No
α_d	Dynamic friction	[0.45, 0.55]	No
D_c	Critical slip	[0.2, 0.8] m	No
σ_n	Normal stress	120 MPa	Yes
c_s	Shear wave speed	3.464 km/s	Yes
ρ	Density	2670 kg/m^3	Yes

Validity constraints

1. **Velocity-weakening**:
   $\alpha_d < \alpha_s$
   (friction must decrease with slip)
2. **Nucleation condition**:
   $\tau_0 > \alpha_s \sigma_n$
   (background stress must exceed peak static friction strength)

LHS oversamples by 3× and rejects invalid draws. All 300 samples generated without issues.